



GarmentX: Autoregressive parametric representations for 3D garment generation

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ABSTRACT

This work presents GarmentX, a novel framework for generating diverse, high-fidelity, and wearable 3D garments from a single input image. Traditional methods directly predict 2D pattern edges and their connectivity, an overly unconstrained approach that often leads to severe self-intersections and physically implausible garment structures. In contrast, GarmentX introduces a structured parametric representation compatible with GarmentCode, ensuring that the decoded sewing patterns always form valid, simulation-ready 3D garments while allowing for intuitive modifications of garment shape and style. To achieve this, we employ a masked autoregressive model that sequentially predicts garment parameters, leveraging autoregressive modeling for structured generation while mitigating inconsistencies in direct pattern prediction. Additionally, we introduce GarmentX dataset, containing 378,682 garment parameter–image pairs, constructed through an automatic data generation pipeline that synthesizes diverse garment images conditioned on parametric garment representations. Through integrating our method with GarmentX dataset, we achieve state-of-the-art performance in geometric fidelity and input image alignment, with 100% simulation success that generalizes across diverse body shapes and poses. The complete source code, pre-trained model weights, and the dataset will be released upon publication.

1. Introduction

3D garments are essential assets for digital fashion, virtual try-on, film VFX, game development, and AR/VR. Beyond visual realism, downstream applications further require the garments to be wearable, i.e. topologically consistent, simulation-ready, and drapable onto a human body. Traditional pipelines meet this bar only by manual modeling in professional software such as CLO3D or by high-fidelity scanning (3dMD, Artec3D), both of which demand specialized expertise or costly hardware and are hard to scale.

Recent deep learning methods attempt to generate 3D garments directly from a single image, but none consistently produces wearable outputs. Pattern-based methods—SewFormer [1], PanelFormer [2], Alpparel [3], ChatGarment [4]—predict 2D sewing-pattern edges and stitches in an unconstrained space, often yielding self-intersecting or unsimulatable panels. Deformation-based methods—Garment3DGen [5],

DrapeNet [6], GarVerseLOD [7]—warp a predefined 3D template and are therefore fragile on out-of-distribution styles. Generic image-to-3D generators—InstantMesh [8], LGM [9], Trellis [10], CLAY [11], unified NeRF-mesh optimization [12], edge-guided multi-view reconstruction [13]—match the input visually but produce closed or double-layered meshes that cannot be worn. The common root cause is that all of them predict garments at the wrong level of abstraction: the output space of raw pattern edges, vertex offsets, or free-form mesh coordinates is vastly larger than the subspace of physically valid garments, so unconstrained prediction almost inevitably lands outside it.

We therefore propose GarmentX, which predicts garments at a higher level of abstraction. Each garment is represented by a compact vector of semantically meaningful attributes—neckline type, sleeve length, skirt flare, and so on—directly compatible with the GarmentCode [14] procedural generator, so that every valid parameter vector

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deterministically decodes into a topologically correct, simulation-ready sewing pattern. Invalid panels, stitching errors, and simulation divergence are eliminated by construction, and the parametric structure additionally enables intuitive single-attribute editing. To predict such a structured vector from an image—where parameters are unordered, mix continuous and discrete types, and exhibit inherent ambiguity under occlusion or visually similar configurations—GarmentX adopts a masked autoregressive (MAR) backbone augmented with a per-token diffusion head, which factorizes the joint distribution into simpler marginals while robustly modeling such ambiguity for continuous parameters.

Training such a model also requires large-scale paired data, which existing datasets do not provide. We therefore build the GarmentX dataset, 378,682 parameter–image pairs produced by an automatic, quality-gated pipeline that (i) maps GarmentCodeData [15] raw attributes into our structured representation and (ii) uses a canny-conditioned ControlNet [16] to render photorealistic images from each vector. Across public benchmarks, GarmentX achieves state-of-the-art geometric fidelity and image alignment with a 0% simulation failure rate. Notably, these gains stem from three complementary pillars: the parametric representation ensures simulation validity and editability by construction, the masked autoregressive model is responsible for geometric fidelity, and the large-scale dataset provides the supervision that makes accurate prediction possible. In summary, our contributions are:

- A structured parametric representation compatible with GarmentCode, guaranteeing simulation-ready and editable garments by construction.
- A masked autoregressive framework with a per-token diffusion head that predicts garment parameters from a single image, achieving state-of-the-art geometric fidelity.
- An automatic, quality-gated data generation pipeline and the resulting GarmentX dataset of 378,682 parameter–image pairs, providing large-scale diverse supervision for training the predictor.

2. Related work

2.1. Open surface and garment pattern generation

Open-surface generation has advanced through unsigned distance fields (UDFs) [17,18], but for clothing, sewing patterns provide a more structured alternative that captures garment topology precisely without UDFs' difficult gradient predictions. Building on the observation [19] that 3D garments decompose into flat 2D panels, pattern-based methods predict panel and stitch data and then simulate. Representative work includes NeuralTailor's [20] LSTM reconstruction, the transformers of SewFormer [1] and PanelFormer [2] for image-to-pattern conversion, and DressCode's [21] GPT-based text-conditioned pattern generation. These methods encode a panel by its rotation, translation, vertex coordinates, edge connectivity, edge types, and curvature, with stitches as one-to-one edge connections. While effective for simple garments, this token sequence explodes for complex designs; the predicted patterns often mismatch the input, with self-intersections or incorrect stitching that cause simulation failures. Recent work mitigates this via large multimodal models [4] or novel tokenization [3], but still predicts the same low-level pattern quantities. Our insight is different: we predict higher-level garment parameters, which are more intuitive, easier to obtain from images, and guarantee valid patterns by construction. All image-conditioned methods above rely on strong visual features; advances in part-aware representation learning [22], domain-adaptive recognition [23], and synthetic-to-real generalization [24] have steadily raised the quality of such features, motivating our use of DINOv2 patch tokens as conditioning.

2.2. Autoregressive models for 3D generation

Autoregressive transformers—pioneered by self-attention [25] and masked autoencoders [26]—have transformed visual generation: early works such as VQGAN [27] and Muse [28] demonstrated the value of discrete tokenization for image synthesis, and this paradigm has since been extended to 3D meshes [29] and scalable text-to-image generation [30]. The common theme is to decompose a complex generation task into a sequence of simpler token predictions, which yields coherent outputs and controllable inference. The idea has also spread to 3D: SAR3D [31] develops multi-scale quantization on top of a VAR backbone, while MAR-3D combines autoregression with diffusion for high-resolution mesh generation. Structured parameter prediction via diffusion has likewise been explored in related domains: DI-PCG [32] for inverse procedural content generation, AniFaceDiff [33] for animating stylized avatars, and cross-modal alignment [34] for bridging visual and structural modalities. Building on these advances, GarmentX applies masked autoregression with a diffusion loss [35] to the space of garment parameters, which are unordered and mix continuous and discrete types—properties the autoregressive and diffusion combination is particularly suited to model.

3. Method

GarmentX is a parametric-driven framework for generating diverse and simulation-ready 3D garments from an input image. It consists of a structured parametric representation that guarantees physical validity (Section 3.1), a large, quality-controlled dataset that aligns images with garment parameters (Section 3.2), and a masked autoregressive model that decodes an input image into such a vector (Section 3.3). An overview is given in Fig. 1.

3.1. GarmentX parameter representation

Instead of predicting low-level geometric entities (pattern edges, stitching rules), GarmentX predicts a compact vector of high-level, semantically meaningful parameters. Such a vector is deterministically decoded into a 3D garment by GarmentCode: a rule-based generator assembles modular component programs (e.g., sleeves, collars, skirts, waistbands) into a 2D sewing pattern, which is then draped onto a body mesh by the NVIDIA Warp XPBD cloth simulator. Both stages depend only on the parameter vector, so predicting a valid vector is sufficient to guarantee a simulation-ready 3D garment.

Formally, a garment is represented by a vector $p = \{p_0, p_1, \dots, p_N\}$ where each p_i is an editable attribute such as neckline type, sleeve shape, pants length, or skirt flare. p contains 125 parameters—79 continuous (e.g., shirt length, sleeve length, skirt flare angle) and 46 ordinal discrete (e.g., 4 collar components). Since these parameters mix types and span very different numerical ranges, we normalize them to a common range $[-1, 1]$ that is directly compatible with our diffusion-based generator.

Normalization. Continuous parameters are linearly projected via

$$x_i = 2 \cdot \frac{p_i - p_{i,\min}}{p_{i,\max} - p_{i,\min}} - 1, \quad (1)$$

where $p_{i,\min}$ and $p_{i,\max}$ are the min and max of the i th parameter. For a discrete attribute with K ordinal choices, we uniformly partition $[-1, 1]$ into K bins and encode the k th choice as the bin midpoint:

$$x_i = \frac{2(k + 0.5)}{K} - 1, \quad k \in \{0, 1, \dots, K - 1\}. \quad (2)$$

For instance, when $K = 3$ the three choices are mapped to $\{-0.667, 0, 0.667\}$. We denote the overall normalization as $x = \mathbb{N}(p)$.

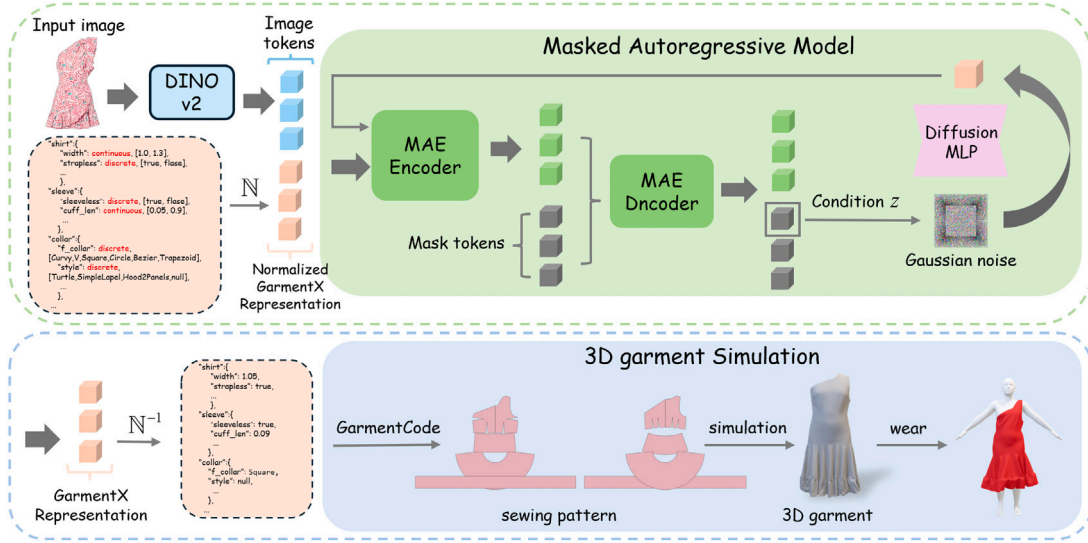


Fig. 1. Overview of the GarmentX pipeline. Given a single image, GarmentX extracts conditioning tokens with DINOv2, mixes them with 125 learnable parameter tokens through an MAE encoder–decoder with random masking, and denoises each parameter via a per-token diffusion MLP. GarmentCode decodes the predicted vector into a sewing pattern and drapes it onto an arbitrary body.

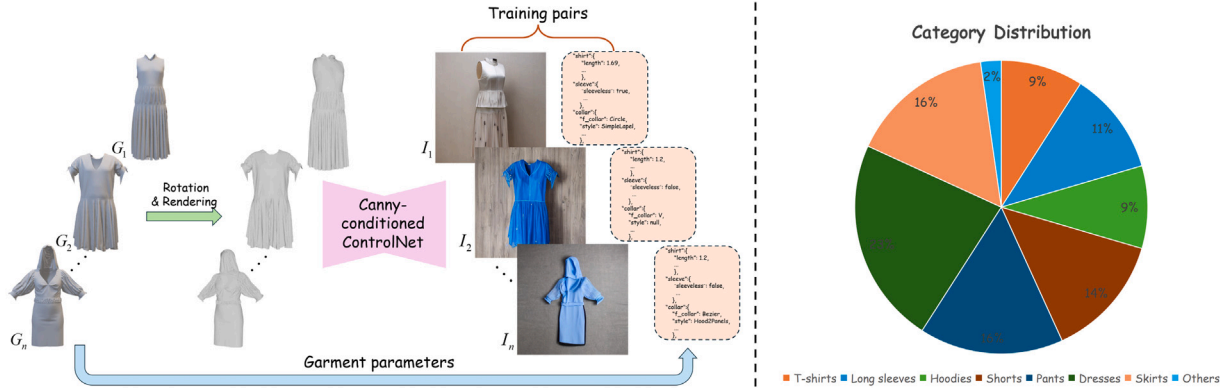


Fig. 2. Automatic data construction pipeline and dataset statistics. Left: we build garment parameter–image pairs with GarmentCodeData, Blender, and a canny-conditioned ControlNet, gated by Canny Edge IoU > 95%. Right: category distribution of the resulting GarmentX dataset.

Denormalization. At inference, the network predicts $\hat{x} \in [-1, 1]$. Continuous values are recovered by inverting Eq. (1); discrete values are snapped to the nearest bin midpoint:

$$k^* = \arg \min_k |\hat{x}_i - m_k|, \quad m_k = \frac{2(k + 0.5)}{K} - 1, \quad (3)$$

with out-of-range predictions clamped to $[-1, 1]$ (e.g., at $K=3$ the midpoints are $\{-0.667, 0, 0.667\}$ and $\hat{x}_i=0.2$ is decoded as the second category).

3.2. GarmentX dataset construction

Training requires large-scale garment images paired with structured parameter vectors, which existing datasets do not provide. We therefore build the GarmentX dataset of 378,682 parameter–image pairs.

The pipeline starts from GarmentCodeData [15], whose records are hierarchically nested YAML files whose leaves are exactly the 125 GarmentCode parameters but store categorical values as strings and use inconsistent numerical ranges. We flatten each record into a single numerical vector by encoding the categorical strings as integer indices and applying the type-aware normalization of Section 3.1; this step is lossless.

As illustrated in Fig. 2(left), the construction pipeline converts each GarmentCodeData record into realistic parameter–image pairs

in four stages. First, each mesh is rendered in Blender from nine viewpoints that combine three azimuths $\{0, 30, -30\}$ with three elevations $\{0, 30, -30\}$, covering front, three-quarter, and slightly overhead or underhead views. Next, every render is passed through a canny-conditioned ControlNet [16] to close the domain gap with real-world photographs, using a fixed prompt template across all categories—base prompt “A modern garment”, additional prompt “best quality, extremely detailed”, and a standard negative prompt that discourages anatomy artifacts. Since ControlNet occasionally hallucinates anatomical details absent from the input mesh, such as legs below a pair of pants or arms attached to a standalone shirt, we apply a canny edge IoU quality gate: edge maps are extracted from both the Blender render and the corresponding ControlNet output, and any pair whose edge-map IoU falls below 95% is discarded; on 70,000 verified pairs the curated data attains a 96.7% mean IoU, confirming that the gate preserves silhouettes faithfully. The resulting GarmentX dataset contains 378,682 parameter–image pairs covering all major clothing categories, whose full distribution is shown in Fig. 2(right).

3.3. 3D garment generation

Given a single input image, our model predicts the 125-dim parameter vector best reconstructing the depicted garment. It has three

components—an image encoder, an MAE [26] encoder–decoder that mixes image tokens with parameter tokens under random masking, and a per-token diffusion head—followed by the GarmentCode decoding of Section 3.1.

Image conditioning. A 224×224 input is fed through DINOv2 [36] ViT-B/14, whose last-layer patch features yield 16×16=256 tokens of dimension 768. These serve as conditioning tokens for the MAE encoder. To enable classifier-free guidance (CFG) at inference, we randomly drop the conditioning tokens with probability 0.1 during training [37].

Masked autoencoder (MAE). The 256 conditioning tokens are concatenated with 125 learnable parameter tokens. During training, a random 70%–100% subset of the parameter tokens is masked using a truncated-Gaussian mask ratio following MAR [35]. The 256 DINOv2 image tokens are never masked. The MAE encoder processes the unmasked tokens and its outputs are concatenated with learnable mask tokens and passed to the MAE decoder, which outputs a per-position conditioning vector $\mathbf{z} \in \mathbb{R}^{768}$ for every masked parameter.

Per-token diffusion head. For each masked position, $\mathbf{z}$ feeds a lightweight diffusion MLP that denoises Gaussian noise into the clean parameter value. Training minimizes

$$\mathcal{L}(\mathbf{z}, x) = \mathbb{E}_{\mathbf{z}, t} [\|\epsilon - \epsilon_\theta(x^t | t, \mathbf{z})\|^2], \quad (4)$$

where x^t is the noised parameter at step t and ϵ is the added Gaussian noise. A diffusion head is essential because a single input image can correspond to multiple plausible parameter configurations—ambiguity that a unimodal regression loss cannot represent.

Inference with cosine-scheduled autoregression. At inference we generate the full 125-parameter vector in S autoregressive steps. A random generation order is sampled; image tokens and all-masked parameter tokens are passed through the MAE encoder–decoder; the diffusion head then denoises the chosen positions in parallel via DDIM [38] for 100 steps. The number of positions unmasked at step s follows the cosine schedule of MaskGIT [39],

$$N_s = \left\lfloor N \left(\cos \frac{s}{S} - \cos \frac{S-1}{S} \right) \right\rfloor, \quad (5)$$

which unmasks fewer positions early and more later. We use $S = 64$; the ablation in Section 4.5 shows further increases yield only marginal gains.

Classifier-free guidance schedule. We additionally apply CFG:

$$\epsilon = \epsilon_\theta(x_t | t, z_u) + \omega_s \cdot (\epsilon_\theta(x_t | t, z_c) - \epsilon_\theta(x_t | t, z_u)), \quad (6)$$

where z_c and z_u are the conditional and unconditional MAE outputs. Following [35], we use a linearly growing coefficient $\omega_s = s \cdot \lambda_{\text{cfg}}/S$ that ramps up from weak guidance to λ_{cfg} ; Section 4.5 confirms that this schedule outperforms both constant-CFG and no-CFG.

From parameters to 3D garments. The predicted 125-dim vector is de-normalized (Section 3.1), passed through GarmentCode to obtain the 2D sewing pattern, and draped onto an SMPL [40] body via NVIDIA Warp. Because every predicted vector lies in a valid region of the parametric space by construction, the pipeline avoids the self-intersection, stitching, and topology failures of prior pattern-based methods.

4. Experiments

This section validates GarmentX on three fronts: state-of-the-art wearable 3D garments (Section 4.2), perceptual superiority on a user study (Section 4.4), and per-component ablations (Section 4.5).

4.1. Datasets, metrics, and implementation details

Datasets. For the standard garment-only regime, we train on the GarmentX dataset and evaluate quantitatively on CLOTH3D [41] and qualitatively on the Complex Virtual Dressing Dataset (CVDD) [42], which contains diverse fashion garments. For the real-world human-worn regime, we train and evaluate a second variant of GarmentX on ChatGarmentDataset [4], using its official train/test split.

Metrics. We use three groups of metrics. (i) Geometric accuracy, applied to all methods, is measured by Chamfer Distance (CD) and Point-to-Surface distance (P2S). All CD/P2S values are computed in normalized $[-1, 1]^3$ units and scaled by 10^2 for readability (e.g., 4.92 means 4.92×10^{-2} in normalized units). Every garment is centered and its bounding box uniformly scaled to $[-1, 1]^3$ before sampling 20,000 points on the garment-only surface. (ii) Wearability, also applied to all methods, checks whether the output is an open-surface single-layer mesh that passes simulation; for methods producing sewing patterns, we additionally report the simulation failure rate (defined below under Simulation protocol). (iii) Pattern-level metrics, applied only to pattern-based baselines that output sewing patterns, follow SewFormer [1]: Panel L2 (L2 error of panel edge parameters), Rot L2 and Trans L2 (per-panel rotation and translation errors), and #Panel and #Edges (accuracy of predicted panel count and of edge count within each correctly predicted panel). Runtime on a single A100 is reported.

Simulation protocol and failure criteria. All simulations use GarmentCode's default configuration: NVIDIA Warp with an XPBD integrator at 60 FPS, 10 substeps, up to 2400 frames, on the default T-pose/A-pose bodies with self-collision and soft body–cloth contacts enabled. A simulation is declared failed if any of: (a) the simulator crashes or produces NaN; (b) no equilibrium is reached within 2400 frames; (c) body–cloth intersections exceed 35; (d) self-intersections exceed 300; (e) any single frame exceeds 60 s; or (f) total simulation time exceeds 600 s. A garment is wearable if it is an open-surface single-layer mesh passing simulation without any failure; the failure rate is $N_{\text{failed}}/N_{\text{total}}$.

Implementation. Input images are resized to 224×224 and encoded by DINOv2 ViT-B/14. The MAE encoder and decoder each use 16 Transformer layers with 16 heads and hidden dimension 768; the per-token diffusion head is a 6-layer MLP of width 1024. Training runs on 2×A100 80 GB GPUs via Accelerate with global batch size 48 (24 per GPU), AdamW (lr = 10^{-4} , weight decay 0.01, $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$), and a cosine-annealing-with-warm-restarts schedule ($T_0=10$ epochs, $T_{\text{mult}}=2$, min LR 1% of initial); the masking ratio follows MAR [35] as a truncated Gaussian (mean 1.0, std 0.25, clipped at 0.7). Convergence takes about 50 h. At inference, each newly unmasked token is decoded with 100 DDIM denoising steps and a linear CFG coefficient $\omega_s = s \cdot \lambda_{\text{cfg}}/S$ ramping from 1.0 to $\lambda_{\text{cfg}}=3.0$ over $S=64$ AR steps. For dataset synthesis, the ControlNet pipeline of Section 3.2 uses Stable Diffusion v1.5 with the official Canny-conditioned ControlNet [16] at 512×512, 20 DDIM steps, guidance scale 7.5, and Canny thresholds [100, 200]; after the IoU > 95% gate, the curated 378,682 pairs are partitioned 90%/5%/5% into train/val/test. Further details on baseline adaptation are provided in Supplementary Material Section S1.

4.2. Comparisons

We compare against seven baselines in three families. Pattern-based methods (SewFormer [1], ChatGarment [4], DressCode [21]) output sewing patterns and are evaluated on the full metric set: CD, P2S, pattern metrics, and simulation failure rate; DressCode is text-guided and therefore evaluated qualitatively only. Deformation-based and generic image-to-3D methods (Garment3DGen [5], InstantMesh [8], LGM [9], Trellis [10]) output meshes without sewing patterns and are evaluated on CD, P2S, and wearability.

Table 1

Quantitative comparison with image-guided pattern-based methods on CLOTH3D. GarmentX achieves the highest performance across all reported metrics.

Method	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)	Fail.↓	Panel L2↓	Rot L2↓	Trans L2↓	#Panel↑	#Edges↑
SewFormer	9.70	10.14	76.3%	10.57	0.1705	1.693	68.7%	72.5%
ChatGarment	5.44	5.93	18.5%	4.81	0.1183	1.114	88.2%	90.9%
Ours	4.92	4.96	0%	3.85	0.0742	0.727	95.8%	97.3%

Table 2

Quantitative comparison with baselines on CLOTH3D. GarmentX attains the lowest CD/P2S and is the only method that is both wearable and editable.

Method	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)	Wearable	Editable
LGM	9.51	7.68	✗	✗
Garment3DGen	6.34	5.91	✓	✗
InstantMesh	5.99	5.70	✗	✗
Trellis	5.17	5.04	✗	✗
Ours	4.92	4.96	✓	✓

4.2.1. Quantitative comparisons

Against pattern-based methods. Our primary comparison is with the two image-guided pattern-based baselines, which share our setting most closely. Table 1 shows GarmentX achieves the highest performance across all reported metrics: it improves CD/P2S by 0.52/0.97 over the strongest competitor ChatGarment, drops the simulation failure rate from 76.3% (SewFormer) and 18.5% (ChatGarment) to 0%, and achieves near-perfect pattern topology (95.8% #Panel, 97.3% #Edges). The simultaneous gains on geometric and pattern-level metrics show that predicting high-level parameters—rather than raw pattern edges—improves both pattern correctness and mesh quality while eliminating the invalid-pattern failure mode.

Against generic 3D generators. Since these methods do not output sewing patterns, Table 2 replaces simulation and pattern-level metrics with two binary indicators, wearable and editable, alongside CD/P2S. GarmentX attains the lowest CD/P2S and is the only method that is both wearable and editable. Notably, Trellis achieves competitive CD/P2S (5.17/5.04) via dense closed meshes enveloping the target, yet these meshes are neither wearable nor editable—showing that CD/P2S alone cannot distinguish true garments from visually similar but unusable geometry.

Runtime. GarmentX takes 15.2s per garment on one A100, broken down as 5.2s (neural generation) + 1.0s (GarmentCode drafting) + 9.0s (Warp simulation), highlighting the efficiency of the neural component. This is 60× faster than Garment3DGen and 8× faster than SewFormer.

4.2.2. Qualitative comparisons

On CLOTH3D. Fig. 3 shows results on CLOTH3D: GarmentX consistently produces structurally sound garments—single-layer, hole-free, and open at the neck, sleeves, and hem—whereas InstantMesh, LGM, and Trellis output closed or double-layered meshes that cannot be worn.

On CVDD fashion garments. As shown in Fig. 4, the baselines fail in three distinct ways: SewFormer produces geometrically distorted garments because its predicted stitching is flawed; Garment3DGen mismatches unfamiliar silhouettes because it is constrained by a fixed template mesh; and the generic image-to-3D generators LGM and InstantMesh output closed, double-layered meshes that cannot be worn. GarmentX, in contrast, encodes garments natively as parameters and therefore recovers fine semantic details that the baselines miss, including irregular ribbons (a, g), collars (b, d, e), shoulder belts (c), and waistlines (f).

Fig. 5 shows cross-sections of samples: Trellis produces double-layered meshes, and in some cases fully closes T-shirts, while GarmentX yields the single-layer open-structure garments that physical simulation requires.

On sewing-pattern quality. As shown in Fig. 6, SewFormer’s patterns often show irregular arrangements, self-intersections, or extraneous panels that cause simulation failures, whereas GarmentX’s patterns maintain correct stitching by construction.

On real-world human-worn images. Fig. 7 shows ChatGarment exhibits several misalignments—incorrect collar (a, b), missing sleeve components (a), inaccurate skirt length (a), wrong lower-body category (b)—while GarmentX aligns precisely with the input and preserves fine structural details. Fig. 8 further demonstrates GarmentX’s generalization on in-the-wild human-worn photographs with diverse styles, lighting, and texture patterns: from a single photograph, GarmentX recovers the correct silhouette, sleeve style, neckline, and hemline, and the resulting garment drapes successfully onto four distinct bodies—female, male, neutral, and neutral in T-pose. This jointly confirms real-world image generalization and wearability across diverse body shapes and poses, showing that the 0% failure rate is not tied to any single canonical body. Quantitative multi-avatar simulation results across four body types are provided in Supplementary Material Section S4.

On uncommon garments and sketch inputs. Although DressCode takes text rather than image input, we include it as a qualitative reference. Fig. 9(left) shows GarmentX generates uncommon fashion garments—asymmetric one-shoulder dresses, top harnesses—that DressCode fails to produce. Fig. 9(right) further shows GarmentX generalizes to sketch inputs despite never being trained on sketches, preserving precise alignment with the input contours.

4.3. Garment editing

Since every coordinate of our representation is semantically meaningful, editing a garment amounts to modifying a single parameter and re-decoding through GarmentCode. Fig. 10 illustrates a sequence of such single-parameter edits (sleeveless → hood → pant → longer): each step produces a predictable, physically valid change while the rest of the garment is preserved, and all intermediate and final outputs remain simulation-ready.

4.4. User study

We conduct a user study on 40 web-fashion photographs to complement geometric metrics with perceptual judgment. 30 participants each rated all 3 methods (SewFormer, ChatGarment, GarmentX, randomized order) per input on three 5-point Likert scales: Realism, Style Fidelity, and Overall Preference. Table 3 reports mean scores with 95% bootstrap CIs; pairwise paired-*t* tests confirm GarmentX outperforms every baseline on all dimensions ($p < 10^{-3}$). A category-wise breakdown and blinded ranking results (GarmentX ranked first in 83%, 88%, and 91% of cases) are provided in Supplementary Material Section S8.

4.5. Ablation study

We ablate each major design choice under three themes: (i) data and representation—effects of data construction, photoreal images, and representation vs. model; (ii) model architecture—effects of backbone architecture, diffusion head, and image encoder; (iii) inference—effects of CFG scale, and CFG schedule together with AR steps.

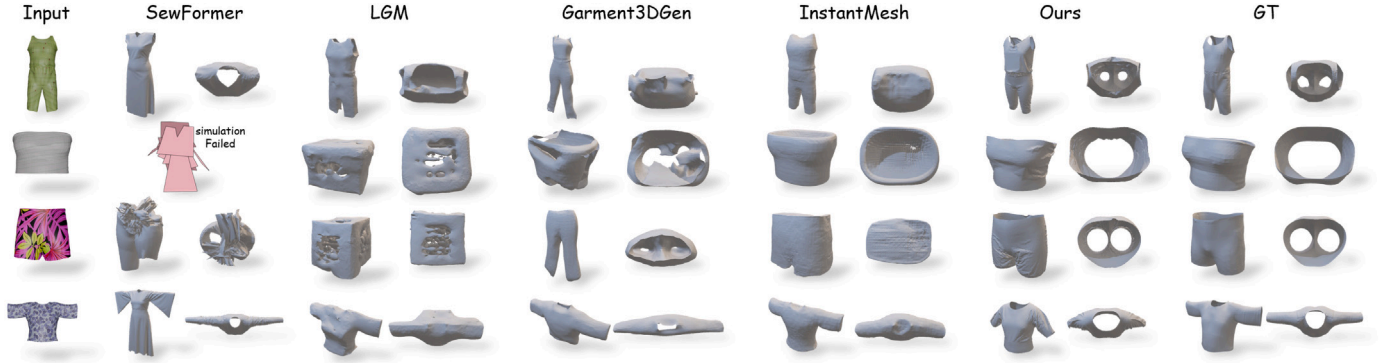


Fig. 3. Qualitative comparisons on CLOTH3D. GarmentX produces single-layer, hole-free garments open at the neck, sleeves, and hem; InstantMesh, LGM, and Trellis produce closed or double-layered meshes.

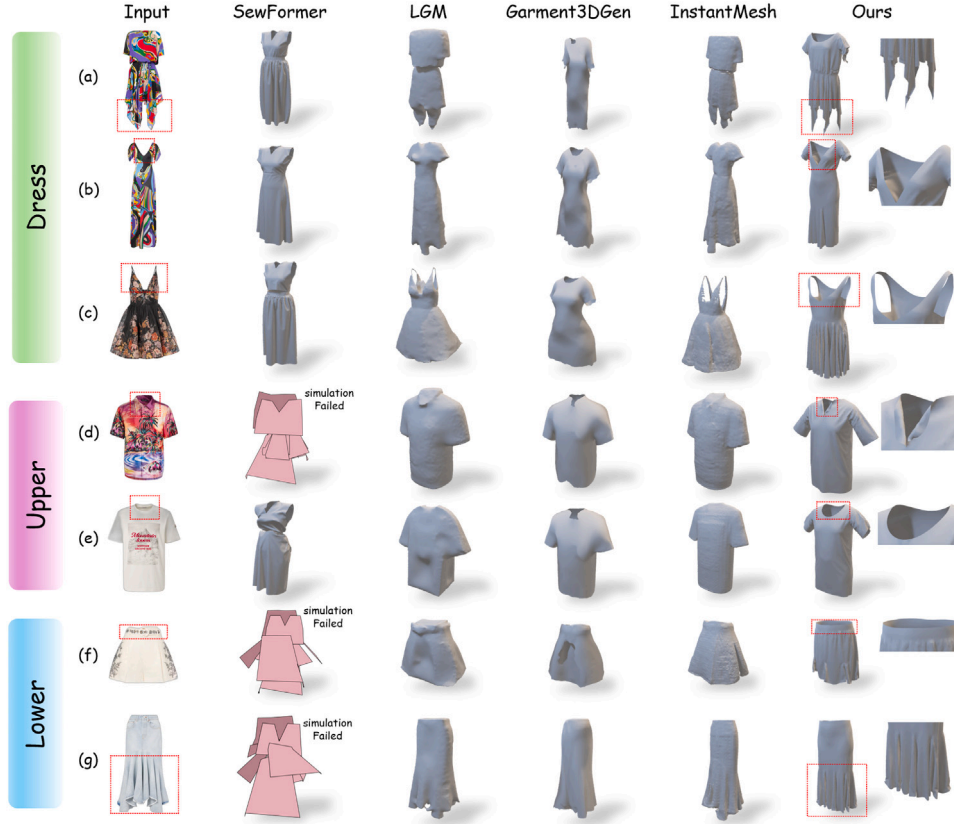


Fig. 4. Qualitative comparisons on CVDD. SewFormer produces self-intersecting panels; Garment3DGen is constrained by its template; LGM and InstantMesh output unwearable closed meshes. GarmentX recovers fine details missed by baselines—ribbons (a, g), collars (b, d, e), shoulder belts (c), and waistlines (f).

Table 3

User study (30 participants, 40 web-fashion images). Means $\pm$ SD with 95% bootstrap CI; all pairwise comparisons favor GarmentX at $p < 10^{-3}$.

Method	Realism $\uparrow$	Style Fidelity $\uparrow$	Overall Preference $\uparrow$
SewFormer	2.87 $\pm$ 0.95 [2.68, 3.06]	2.73 $\pm$ 0.91 [2.55, 2.91]	2.76 $\pm$ 0.93 [2.57, 2.94]
ChatGarment	3.42 $\pm$ 0.82 [3.25, 3.58]	3.29 $\pm$ 0.87 [3.12, 3.47]	3.31 $\pm$ 0.84 [3.14, 3.48]
Ours	4.21 $\pm$ 0.63 [4.08, 4.34]	4.05 $\pm$ 0.71 [3.90, 4.19]	4.15 $\pm$ 0.66 [4.01, 4.28]

4.5.1. Data and representation

Effect of data construction. Two families exist for producing training images from a 3D mesh: render-then-generate (ControlNet, T2I-Adapter [43]) and texture-then-render (Paint3D [44], DreamMat [45]). Texture generation takes 5–20 min per sample and is impractical at our scale, whereas ControlNet runs in 5 s; Fig. 11 further shows ControlNet

produces substantially higher image quality than T2I-Adapter, so we adopt it.

Effect of photoreal images. Training on ControlNet-enhanced images rather than raw Blender renders reduces CD/P2S from 5.84/5.91 to 4.92/4.96, confirming that closing the render-to-real domain gap matters.

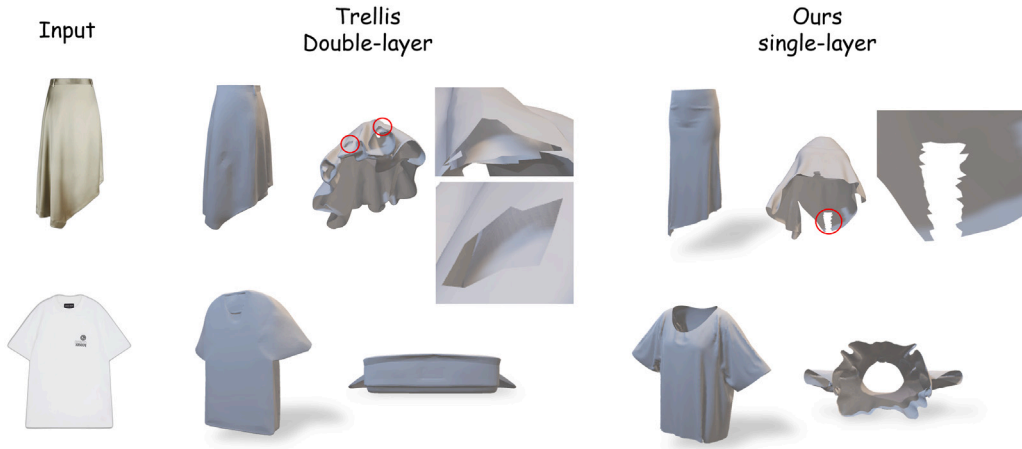


Fig. 5. Cross-sectional views of CVDD samples. Trellis produces double-layered meshes and sometimes fully closes T-shirts, whereas GarmentX produces single-layer, open-structure garments ready for simulation.

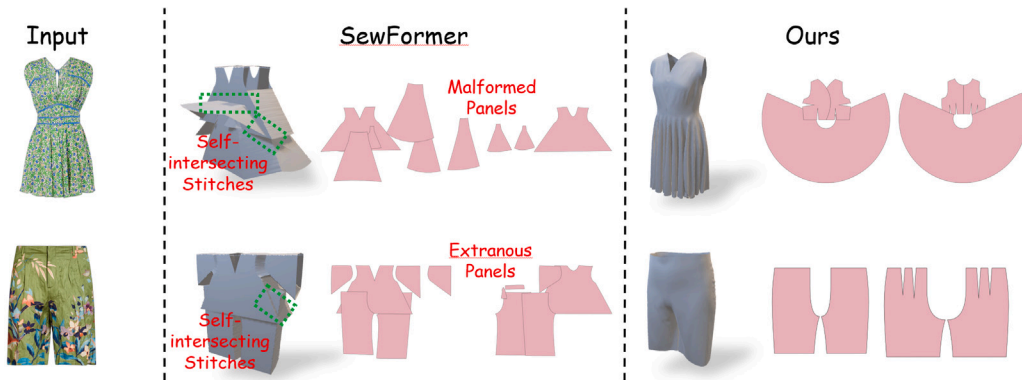


Fig. 6. Pattern-level comparison with SewFormer. SewFormer's patterns are irregular and self-intersecting, causing simulation failures; GarmentX's are correctly arranged and stitched.

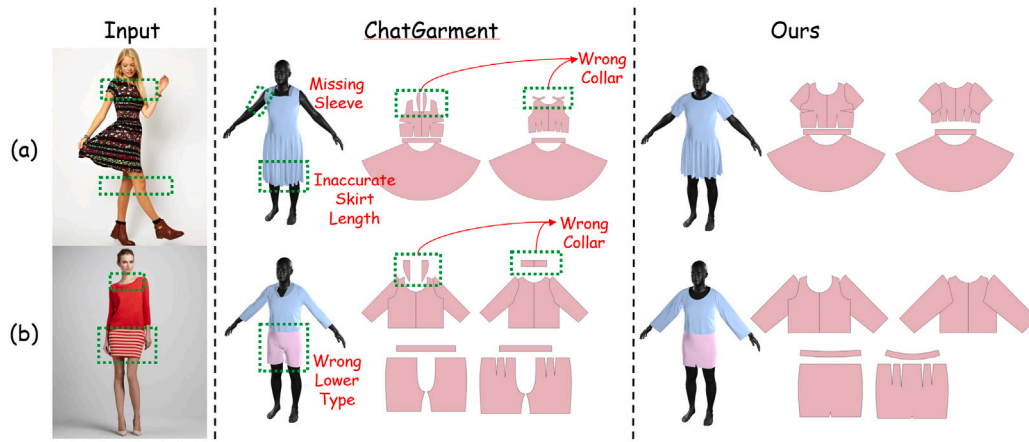


Fig. 7. Pattern-level comparison with ChatGarment. ChatGarment shows collar, sleeve, and length misalignments; GarmentX aligns precisely with the input.

Effect of representation vs. model. A 2×2 factorial crossing two models (GarmentX, SewFormer) with two representations (parametric, raw pattern) in Table 4 isolates two effects: (a) the parametric representation eliminates simulation failures for both models, so 0% failure is a structural property of the representation; (b) under the shared parametric representation, GarmentX still outperforms SewFormer (4.92/4.96 vs. 8.52/8.71), so MAR independently drives mesh quality.

4.5.2. Model architecture

Effect of backbone architecture. Under identical DINOv2 features and a 125-dim target, we benchmark MAR against three lightweight regressors and three DiT [46] sizes. Table 5 shows that MAR attains the best CD/P2S across the entire capacity spectrum. The advantage therefore does not come from capacity, but from MAR's autoregressive factorization and per-token diffusion head. A generation-order sensitiv-

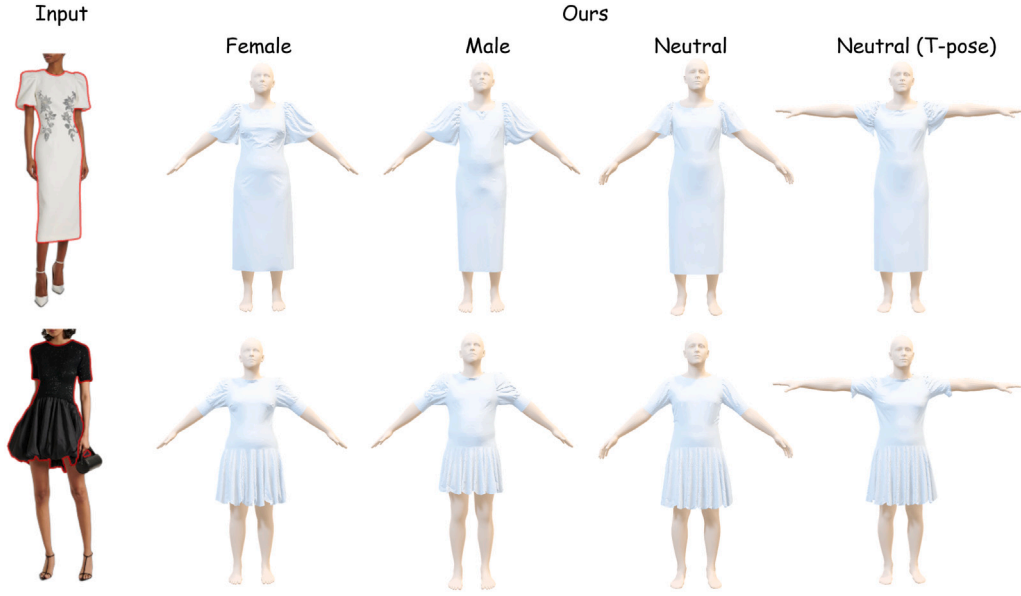


Fig. 8. Real-world generalization and wearability across diverse bodies. From a single in-the-wild photograph, GarmentX recovers the correct silhouette, sleeves, neckline, and hemline, and drapes onto four distinct bodies (female, male, neutral, and neutral in T-pose).

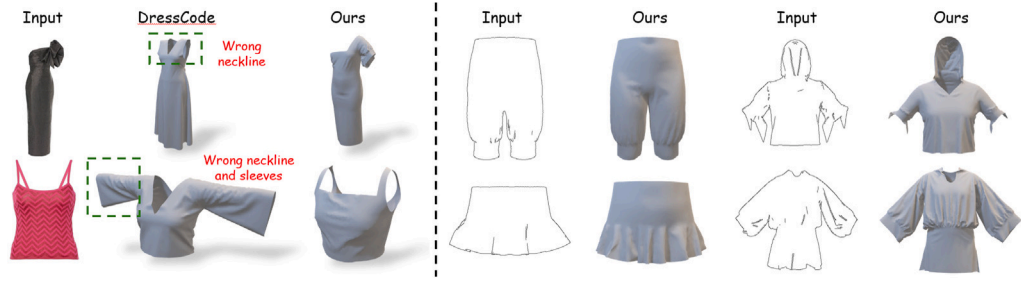


Fig. 9. Qualitative comparison with DressCode and sketch-conditioned generation. Left: GarmentX generates uncommon fashion garments that DressCode fails to produce. Right: Although never trained on sketches, GarmentX produces garments whose silhouettes align precisely with the input contours.

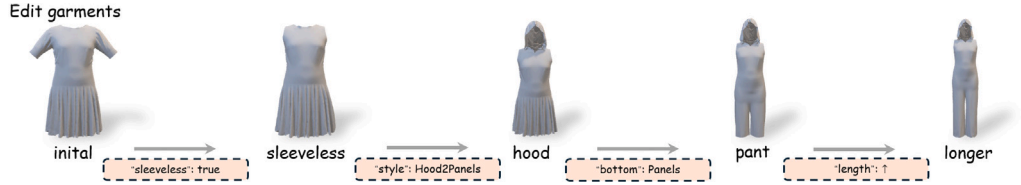


Fig. 10. Sequential single-parameter editing. From a base garment, each step modifies one semantic parameter (sleeveless → hood → pant → longer); intermediate results stay simulation-ready and unchanged regions are preserved.

Table 4

Ablation on representation vs. model. The parametric representation eliminates simulation failures for both models, and under this shared representation MAR further improves CD/P2S over SewFormer.

Model	Representation	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)	Failure (%)↓
SewFormer	Raw Pattern	12.85	12.91	76.3
SewFormer	Parametric	8.52	8.71	0.0
GarmentX	Raw Pattern	7.94	8.03	17.6
GarmentX	Parametric	4.92	4.96	0.0

ity analysis comparing fixed, random, learned, and non-autoregressive variants is provided in Supplementary Material Section S7.

Effect of diffusion head and image encoder. Table 6 ablates two components. Replacing the diffusion head with a direct MSE regressor raises CD/P2S from 4.92/4.96 to 7.83/7.91, because a unimodal loss

Table 5

Ablation on backbone architecture. MAR outperforms lightweight regressors (MLP, ResNet-50/152) and DiT at comparable or larger capacity, showing the gain comes from the masked autoregressive design rather than model size.

Method	Params	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)
MLP (6-layer, dim 256)	0.31M	8.31	8.47
ResNet-50 regressor	25.6M	7.12	7.24
DiT-S	30M	6.47	6.40
ResNet-152 regressor	60.2M	6.68	6.73
DiT-B	119M	6.47	6.35
MAR (Ours)	352M	4.92	4.96
DiT-XL	607M	6.17	6.20

collapses to the mean of a multi-modal distribution that diffusion captures. This is verified by a diversity-under-occlusion experiment



Fig. 11. Ablation on data construction. ControlNet produces higher image quality than T2I-Adapter while running faster than texture-then-render pipelines.

Table 6

Ablation on diffusion head and image encoder. Replacing the diffusion head with a direct MSE regressor, or DINOv2 with CLIP, both degrade CD/P2S.

Component	Setting	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)
Head	Direct regression	7.83	7.91
	Diffusion (Ours)	4.92	4.96
Encoder	CLIP ViT-B/16	5.62	5.68
	DINOv2 ViT-B/14 (Ours)	4.92	4.96

Table 7

Ablation on CFG schedule and AR steps S . A linear schedule with $S=64$ matches the accuracy of $S=128$ at half the compute.

Component	Setting	CD↓ ($\times 10^{-2}$)	P2S↓ ($\times 10^{-2}$)
CFG schedule	No CFG ($\omega_s=1.0$)	6.05	6.12
	Constant CFG ($\omega_s=3.0$)	5.31	5.38
	Linear(1.0 $\rightarrow$ 3.0, Ours)	4.92	4.96
AR steps S	$S=16$	5.74	5.81
	$S=32$	5.12	5.18
	$S=64$ (Ours)	4.92	4.96
	$S=128$	4.89	4.94

(Supplementary Material Section S5) and a discrete-parameter analysis comparing unified diffusion against a mixed continuous–categorical variant (Supplementary Material Section S6). Replacing the DINOv2 ViT-B/14 encoder with CLIP ViT-B/16 degrades CD/P2S to 5.62/5.68, as CLIP’s global pooling provides weaker spatial cues than DINOv2’s patch-level features for geometric regression.

4.5.3. Inference

Effect of CFG scale. Sweeping a constant CFG scale $\lambda_{\text{cfg}} \in \{2.0, 3.0, 4.0, 5.0\}$ on CLOTH3D yields CD/P2S of 5.14/5.21, **4.92/4.96**, 5.37/5.38, and 5.03/5.07 respectively, so quality peaks at $\lambda_{\text{cfg}} = 3.0$ and we adopt this value.

Effect of CFG schedule and AR steps. Table 7 ablates the CFG schedule and the number of AR steps S . The linear schedule $\omega_s = s \cdot \lambda_{\text{cfg}}/S$ outperforms constant 3.0 and no-CFG, confirming that guidance should ramp up as more tokens are revealed. Doubling S from 64 to 128 improves CD only from 4.92 to 4.89 at twice the cost, so we adopt $S = 64$.

5. Conclusion and future work

GarmentX is a parametric framework that predicts GarmentCode-compatible parameters from a single image, yielding physically valid,

editable, and simulation-ready 3D garments by construction. Two complementary factors underpin these results: the parametric representation guarantees 0% simulation failure, and the masked autoregressive model drives geometric fidelity. GarmentX achieves state-of-the-art CD/P2S (4.92/4.96), a 0% failure rate, 15.2 s inference, and the highest preference in a 30-participant user study. A discussion of limitations is provided in Supplementary Material Section S2.

Three directions extend this work. First, differentiable GarmentCode would unlock end-to-end physics-based losses that are currently blocked by the rule-based decoder and XPBD simulator: either a neural surrogate trained on GarmentCode input–output pairs, or a smooth re-implementation of its procedural rules using differentiable conditionals and soft geometric primitives. Second, hybrid representations for decorative elements would cover non-structural details such as bows, ruffles, and embroidery that fall outside our parameter vocabulary: the garment body remains handled by the parametric pipeline (preserving topological validity and simulation readiness), while a complementary module—e.g., 3D Gaussian Splatting [47] or a feed-forward reconstruction model—recovers accessories and composites them at anchor regions such as the collar, waistline, or hemline. Third, material-parameter prediction is a natural extension within the same architecture: GarmentCode already exposes bending stiffness, fabric density, and friction, so jointly sampling geometric and material parameters, simulating, and re-synthesizing images through the existing ControlNet pipeline yields augmented supervision for the same MAR model—turning GarmentX into a predictor of physical appearance, not only geometry.

CRedit authorship contribution statement

Jingfeng Guo: Writing – original draft, Conceptualization. **Jinnan Chen:** Writing – original draft. **Weikai Chen:** Conceptualization. **Zhenyu Sun:** Validation. **Lanjiong Li:** Resources. **Baozhu Zhao:** Data curation. **Lingting Zhu:** Conceptualization. **Xin Wang:** Conceptualization. **Qi Liu:** Writing – review & editing, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors confirm that no generative AI or AI-assisted tools were used at any stage of preparing this manuscript. All text, figures, and analyses are entirely the original work of the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2026.114937>.

Data availability

Data will be made available on request.

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